Small LAN with Internet Access

Networking Checkpoint Report

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1. Network Overview

1.1 Scenario

A company floor network consists of:

1 Router (with Internet access)

2 Switches

4 Employee PCs

1 Network Printer

1 Wi-Fi Access Point

2 Wireless Devices

1.2 Network Topology

```
          [Internet]
              |
          [Router]
          /   |   \
         /    |    \
    [Switch 1]  |  [Switch 2]
     /  |  \    |     |
  PC1 PC2 PC3  [AP]  PC4 + Printer
            /  \
          W1   W2
```

Part 1: MAC & Frame Handling

1.1 MAC Address Assignments

Device	MAC Address
Router (LAN)	00:1A:2B:3C:4D:01
Switch 1	00:1A:2B:3C:4D:10
Switch 2	00:1A:2B:3C:4D:11
PC1	00:1A:2B:3C:4D:21
PC2	00:1A:2B:3C:4D:22
PC3	00:1A:2B:3C:4D:23
PC4	00:1A:2B:3C:4D:24
Printer	00:1A:2B:3C:4D:30
Wi-Fi AP	00:1A:2B:3C:4D:40
Wireless Device 1	00:1A:2B:3C:4D:51
Wireless Device 2	00:1A:2B:3C:4D:52

Each device has a unique 48-bit MAC address used for communication within the local network.

1.2 Ethernet Frame Structure (PC → Printer)

When **PC1** sends data to the **Printer**, the Ethernet frame structure is:

```
+------------+-------------+-------------+--------+----------------+------+
| Preamble   | Dest MAC    | Source MAC  | Type   | Payload        | FCS  |
| (8 bytes)  | (6 bytes)   | (6 bytes)   | (2 B)  | (46 – 1500 B)  | (4 B)|
+------------+-------------+-------------+--------+----------------+------+
```

Example Values (PC1 → Printer)

Field	Value
Destination MAC	00:1A:2B:3C:4D:30
Source MAC	00:1A:2B:3C:4D:21
Type	0x0800 (IPv4)
Payload	IP packet (IP header + TCP/UDP + data)
FCS	CRC-32 checksum

Switch Behavior

- The switch learns MAC addresses from incoming frames.

- It forwards frames only to the correct port.
- Each switch port represents a separate collision domain.

1.3 Collision Handling

Wired Ethernet (Switch-Based)

1. Operates in **Full Duplex**
2. No collisions occur
3. Each port is its own collision domain

Legacy Ethernet (Hub-Based)

Uses **CSMA/CD**:

1. Listen before transmitting.
2. If collision occurs → send jam signal.
3. Wait random backoff time.
4. Retransmit.

Wi-Fi (Wireless)

Uses **CSMA/CA**:

1. Listen before sending.
2. Wait random backoff if busy.
3. Optional RTS/CTS handshake.
4. Receive ACK confirmation.

Wireless cannot detect collisions while transmitting, so it avoids them instead.

Part 2: IP Addressing & Subnetting

2.1 Private Address Range

Chosen network:

192.168.10.0/24

Total available addresses: 256
Usable hosts: 254

2.2 Subnet Design

Subnet	Purpose	CIDR	Usable Hosts
Subnet A	Wired devices	192.168.10.0/25	126
Subnet B	Wireless devices	192.168.10.128/25	126

Note: In small deployments, all devices can also share 192.168.10.0/24.

2.3 IP Address Assignments

Device	IP Address	Subnet Mask	Default Gateway
Router (LAN)	192.168.10.1	255.255.255.128	—
PC1	192.168.10.10	255.255.255.128	192.168.10.1
PC2	192.168.10.11	255.255.255.128	192.168.10.1
PC3	192.168.10.12	255.255.255.128	192.168.10.1
PC4	192.168.10.13	255.255.255.128	192.168.10.1
Printer	192.168.10.20	255.255.255.128	192.168.10.1
Wi-Fi AP	192.168.10.129	255.255.255.128	192.168.10.1
Wireless Device 1	192.168.10.130	255.255.255.128	192.168.10.1
Wireless Device 2	192.168.10.131	255.255.255.128	192.168.10.1

2.4 Role of DHCP

DHCP (Dynamic Host Configuration Protocol):

- Client sends **DHCP Discover**
- Server replies with **DHCP Offer**
- Client sends **DHCP Request**
- Server sends **DHCP Acknowledgment**

DHCP automatically provides:

- IP address
- Subnet mask
- Default gateway
- DNS server

Benefits:

Reduces configuration errors

Simplifies network management

Enables automatic device setup

Part 3: Routing & Packet Forwarding

3.1 Scenario

Wireless Device 1 (192.168.10.130) sends a request to:

https://www.example.com

3.2 Packet Path Overview

W1 → Wi-Fi AP → Router → ISP → Internet → Web Server

3.3 Step-by-Step Packet Flow

Step	Action
1	W1 builds IP packet (Src=192.168.10.130, Dst=Public IP).
2	Frame sent to AP (destination MAC = AP).
3	AP forwards to Router.
4	Router checks routing table.
5	Router performs NAT translation.
6	Packet forwarded to ISP gateway.
7	Internet routers forward based on destination IP.
8	Server replies to router's public IP.
9	Router uses NAT table to forward reply to W1.

3.4 IP Header Fields Used by Routers

Routers inspect:

1. Destination IP address

2. TTL (decremented by 1)
3. Header checksum (recalculated)

They do not modify IP addresses except when performing NAT.

3.5 Routing Table Example

Destination	Next Hop	Interface
192.168.10.0/24	—	LAN
0.0.0.0/0	ISP Gateway	WAN

Routers choose the **longest prefix match** to determine the next hop.

3.6 Network Address Translation (NAT)

Problem:

Private IPs (192.168.x.x) are not routable on the Internet.

Solution:

Router replaces private source IP with its public WAN IP.

Maintains NAT table mapping:

(192.168.10.130 : port) ↔ (203.0.113.5 : port)

Result:

Multiple internal devices share one public IP.

Replies return to correct internal device.

Final Summary

This network design demonstrates:

Proper MAC addressing and frame forwarding

Collision handling in wired and wireless networks

Structured IP addressing and subnetting

DHCP-based automatic configuration

Router-based packet forwarding

NAT enabling Internet access for private networks